

# Machine Learning

## Basics of Classification

**FAST** 

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DISCOVERING  
THE FUTURE

# Topics of previous lecture

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics
- ✓ Basic Linear Classifier

# Topics of today's lecture

- K-Nearest Neighbours Classifier
- Naive Bayes Classifier

# Tennis dataset

| Day | Outlook  | Temp | Humidity | Wind   | PlayTennis |
|-----|----------|------|----------|--------|------------|
| D1  | Sunny    | Hot  | High     | Weak   | No         |
| D2  | Sunny    | Hot  | High     | Strong | No         |
| D3  | Overcast | Hot  | High     | Weak   | Yes        |
| D4  | Rain     | Mild | High     | Weak   | Yes        |
| D5  | Rain     | Cool | Normal   | Weak   | Yes        |
| D6  | Rain     | Cool | Normal   | Strong | No         |
| D7  | Overcast | Cool | Normal   | Strong | Yes        |
| D8  | Sunny    | Mild | High     | Weak   | No         |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes        |
| D10 | Rain     | Mild | Normal   | Weak   | Yes        |
| D11 | Sunny    | Mild | Normal   | Strong | Yes        |
| D12 | Overcast | Mild | High     | Strong | Yes        |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes        |
| D14 | Rain     | Mild | High     | Strong | No         |

# Tennis dataset

| Day | Outlook  | Temp                                                                                                                             | Humidity | Wind   | PlayTennis |
|-----|----------|----------------------------------------------------------------------------------------------------------------------------------|----------|--------|------------|
| D1  | Sunny    | Hot                                                                                                                              | High     | Weak   | No         |
| D2  | Sunny    | Hot                                                                                                                              | High     | Strong | No         |
| D3  | Overcast | Hot                                                                                                                              | High     | Weak   | Yes        |
| D4  | Rain     | Mild                                                                                                                             | High     | Weak   | Yes        |
| D5  | Rain     | Cool                                                                                                                             | Normal   | Weak   | Yes        |
| D6  | Rain     | Let's leave these features out for now: <ul style="list-style-type: none"><li>• Temp</li><li>• Humidity</li><li>• Wind</li></ul> |          |        |            |
| D7  | Overcast |                                                                                                                                  |          |        |            |
| D8  | Sunny    |                                                                                                                                  |          |        |            |
| D9  | Sunny    |                                                                                                                                  |          |        |            |
| D10 | Rain     |                                                                                                                                  |          |        |            |
| D11 | Sunny    |                                                                                                                                  |          |        |            |
| D12 | Overcast | Mild                                                                                                                             | High     | Strong | Yes        |
| D13 | Overcast | Hot                                                                                                                              | Normal   | Weak   | Yes        |
| D14 | Rain     | Mild                                                                                                                             | High     | Strong | No         |

# Tennis dataset

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D3  | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D7  | Overcast | Yes        |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D10 | Rain     | Yes        |
| D11 | Sunny    | Yes        |
| D12 | Overcast | Yes        |
| D13 | Overcast | Yes        |
| D14 | Rain     | No         |

# Tennis dataset

Now consider the classification task of predicting label **PlayTennis** from a single feature **Outlook**

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D1  | Sunny    | No         |
|     | Sunny    | No         |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |
|     | Overcast | Yes        |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D10 | Rain     | Yes        |
| D11 | Sunny    | Yes        |
| D12 | Overcast | Yes        |
| D13 | Overcast | Yes        |
| D14 | Rain     | No         |

# Tennis dataset

Now consider the classification task of predicting label **PlayTennis** from a single feature **Outlook**

First, let's sort the rows by **Outlook**

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D1  | Sunny    | No         |
|     | Sunny    | No         |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |
|     | Overcast | Yes        |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
|     | Rain     | Yes        |
|     | Sunny    | Yes        |
| D12 | Overcast | Yes        |
| D13 | Overcast | Yes        |
| D14 | Rain     | No         |

# Tennis dataset

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
| D7  | Overcast | Yes        |
| D12 | Overcast | Yes        |
| D13 | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

Can we apply some learning algorithm that we already know?

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

Can we apply some learning algorithm that we already know?

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
|     | Sunny    | No         |
|     | Sunny    | No         |
|     | Sunny    | No         |
|     | Sunny    | Yes        |
|     | Sunny    | Yes        |

No! Linear classifiers need numeric features! We must transform the data into numeric first!

# Tennis dataset

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
| D7  | Overcast | Yes        |
| D12 | Overcast | Yes        |
| D13 | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

Is this a good encoding?

- 0=Overcast
- 1=Sunny
- 2=Rain

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |

Is this a good encoding?

- 0=Overcast
- 1=Sunny
- 2=Rain

No, it's not a good encoding!

No combination of weights results in predicting *PlayTennis=Yes* for *Overcast* and *Rain* and *PlayTennis=No* for *Sunny*.

If the weight for *Outlook* is positive, then changing *Outlook* from *Overcast*  $\rightarrow$  *Sunny*  $\rightarrow$  *Rain* increases the total score (and if weight is negative then decreases).

# Better encoding (1-hot encoding)

| Day | Outlook=Overcast | Outlook=Rain | Outlook=Sunny | PlayTennis |
|-----|------------------|--------------|---------------|------------|
| D3  | 1                | 0            | 0             | Yes        |
| D7  | 1                | 0            | 0             | Yes        |
| D12 | 1                | 0            | 0             | Yes        |
| D13 | 1                | 0            | 0             | Yes        |
| D4  | 0                | 1            | 0             | Yes        |
| D5  | 0                | 1            | 0             | Yes        |
| D6  | 0                | 1            | 0             | No         |
| D10 | 0                | 1            | 0             | Yes        |
| D14 | 0                | 1            | 0             | No         |
| D1  | 0                | 0            | 1             | No         |
| D2  | 0                | 0            | 1             | No         |
| D8  | 0                | 0            | 1             | No         |
| D9  | 0                | 0            | 1             | Yes        |
| D11 | 0                | 0            | 1             | Yes        |

# Better encoding (1-hot encoding)

| Day | Outlook=Overcast | Outlook=Rain | Outlook=Sunny | PlayTennis |
|-----|------------------|--------------|---------------|------------|
| D3  | 1                | 0            | 0             | Yes        |
| D7  | 1                | 0            | 0             | Yes        |
| D12 | 1                | 0            | 0             | Yes        |
| D13 | 1                | 0            | 0             | Yes        |
| D4  | 0                |              |               | Yes        |
| D5  | 0                |              |               | Yes        |
| D6  | 0                |              |               | No         |
| D10 | 0                |              |               | Yes        |
| D14 | 0                | 1            | 0             | No         |
| D1  | 0                | 0            | 1             | No         |
| D2  | 0                | 0            | 1             | No         |
| D8  | 0                | 0            | 1             | No         |
| D9  | 0                | 0            | 1             | Yes        |
| D11 | 0                | 0            | 1             | Yes        |

Now for linear models there would be a separate weight for each value of *Outlook*

# One-hot encoding

- One-hot encoding: Introduce a binary (1/0) variable for each possible value of the categorical variable
- Using one-hot encoding is a standard method of transforming a nominal feature (unordered categorical feature) into numeric features
- For ordinal features (ordered categorical features, e.g. grades) it can also be used, but it would ignore the ordering

# Are the 2 classes linearly separable?

| Day | Outlook=Overcast | Outlook=Rain | Outlook=Sunny | PlayTennis |
|-----|------------------|--------------|---------------|------------|
| D3  | 1                | 0            | 0             | Yes        |
| D7  | 1                | 0            | 0             | Yes        |
| D12 | 1                | 0            | 0             | Yes        |
| D13 | 1                | 0            | 0             | Yes        |
| D4  | 0                | 1            | 0             | Yes        |
| D5  | 0                | 1            | 0             | Yes        |
| D6  | 0                | 1            | 0             | No         |
| D10 | 0                | 1            | 0             | Yes        |
| D14 | 0                | 1            | 0             | No         |
| D1  | 0                | 0            | 1             | No         |
| D2  | 0                | 0            | 1             | No         |
| D8  | 0                | 0            | 1             | No         |
| D9  | 0                | 0            | 1             | Yes        |
| D11 | 0                | 0            | 1             | Yes        |

# Are the 2 classes linearly separable?

| Day | Outlook=Overcast | Outlook=Rain | Outlook=Sunny | PlayTennis |
|-----|------------------|--------------|---------------|------------|
| D3  | 1                | 0            | 0             | Yes        |
| D7  | 1                | 0            | 0             | Yes        |
| D12 | 1                | 0            | 0             | Yes        |
| D13 | 1                | 0            | 0             | Yes        |
| D4  | 0                | 1            | 0             | Yes        |
| D5  | 0                | 1            | 0             | Yes        |
| D6  | 0                | 1            | 0             | No         |
| D10 | 0                | 1            | 0             | Yes        |
| D14 | 0                | 1            | 0             | No         |
| D1  | 0                | 0            | 1             | No         |
| D2  | 0                | 0            | 1             | No         |
| D8  | 0                | 0            | 1             | No         |
| D9  | 0                | 0            | 1             | Yes        |
| D11 | 0                | 0            | 1             | Yes        |

Not linearly separable!  
For example, look at **D6, D10**  
instances that have the same features  
but belong to different classes.

# Tennis dataset

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
| D7  | Overcast | Yes        |
| D12 | Overcast | Yes        |
| D13 | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

If *Outlook*=*Rain*,  
what would you  
predict?

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

If *Outlook*=*Rain*,  
what would you  
predict?

If *Outlook*=*Rain*,  
what would be the  
probability of  
*PlayTennis*=*Yes*?

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |
|     | Rain     | Yes        |
|     | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

If *Outlook*=*Rain*,  
what would you  
predict?

If *Outlook*=*Rain*,  
what would be the  
probability of  
*PlayTennis*=*Yes*?

What kind of  
probability is it?

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |
|     | Rain     | Yes        |
|     | Rain     | No         |
|     | Sunny    | No         |
|     | Sunny    | No         |
|     | Sunny    | No         |
|     | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Tennis dataset

If *Outlook*=*Rain*,  
what would you  
predict?

If *Outlook*=*Rain*,  
what would be the  
probability of  
*PlayTennis*=*Yes*?

What kind of  
probability is it?

**Conditional Probability!**

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |
|     | Rain     | Yes        |
|     | Rain     | No         |
|     | Sunny    | No         |
|     | Sunny    | No         |
|     | Sunny    | No         |
|     |          | Yes        |
| D11 | Sunny    | Yes        |

# Conditional Probability

| Day | Outlook  | PlayTennis |  | Day | Outlook | PlayTennis |
|-----|----------|------------|--|-----|---------|------------|
| D3  | Overcast | Yes        |  | D6  | Rain    | No         |
| D7  | Overcast | Yes        |  | D10 | Rain    | Yes        |
| D12 | Overcast | Yes        |  | D14 | Rain    | No         |
| D13 | Overcast | Yes        |  | D1  | Sunny   | No         |
| D4  | Rain     | Yes        |  | D2  | Sunny   | No         |
| D5  | Rain     | Yes        |  | D8  | Sunny   | No         |
| D9  | Sunny    | Yes        |  | D11 | Sunny   | Yes        |

Our intuition tells us to consider the following conditional probability.

$$P(\text{PlayTennis} = \text{Yes} | \text{Outlook} = \text{Rain}) =$$

# Conditional Probability

| Day | Outlook  | PlayTennis |  | Day | Outlook | PlayTennis |
|-----|----------|------------|--|-----|---------|------------|
| D3  | Overcast | Yes        |  | D6  | Rain    | No         |
| D7  | Overcast | Yes        |  | D10 | Rain    | Yes        |
| D12 | Overcast | Yes        |  | D14 | Rain    | No         |
| D13 | Overcast | Yes        |  | D1  | Sunny   | No         |
| D4  | Rain     | Yes        |  | D2  | Sunny   | No         |
| D5  | Rain     | Yes        |  | D8  | Sunny   | No         |
| D9  | Sunny    | Yes        |  | D11 | Sunny   | Yes        |

Our intuition tells us to consider the following conditional probability.

$$\begin{aligned} P(\text{PlayTennis} = \text{Yes} | \text{Outlook} = \text{Rain}) &= \\ &= \frac{P(\text{PlayTennis} = \text{Yes}, \text{Outlook} = \text{Rain})}{P(\text{Outlook} = \text{Rain})} = \end{aligned}$$

# Conditional Probability

| Day | Outlook  | PlayTennis |  | Day | Outlook | PlayTennis |
|-----|----------|------------|--|-----|---------|------------|
| D3  | Overcast | Yes        |  | D6  | Rain    | No         |
| D7  | Overcast | Yes        |  | D10 | Rain    | Yes        |
| D12 | Overcast | Yes        |  | D14 | Rain    | No         |
| D13 | Overcast | Yes        |  | D1  | Sunny   | No         |
| D4  | Rain     | Yes        |  | D2  | Sunny   | No         |
| D5  | Rain     | Yes        |  | D8  | Sunny   | No         |
| D9  | Sunny    | Yes        |  | D11 | Sunny   | Yes        |

Our intuition tells us to consider the following conditional probability.

$$\begin{aligned} P(\text{PlayTennis} = \text{Yes} | \text{Outlook} = \text{Rain}) &= \\ = \frac{P(\text{PlayTennis} = \text{Yes}, \text{Outlook} = \text{Rain})}{P(\text{Outlook} = \text{Rain})} &= \frac{\frac{3}{14}}{\frac{5}{14}} = \frac{3}{5} = 0.6 \end{aligned}$$

# Probabilistic Assumptions

- In classification, it is typically assumed that:
  - There is some unknown process that creates labelled instances for us
  - Each instance is created independently from others, by the same process
  - In statistical terms: we assume that the instances are i.i.d. (independent and identically distributed)

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- In previous calculations we have assumed even more:
  - The sample space (set of all possible outcomes) is the given dataset
  - In other words, the process which creates instances is just selecting a random row from the given dataset

# Probabilistic Assumptions

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- In previous calculations we have assumed even more:
  - The sample space (set of all possible outcomes) is the given dataset
  - In other words, the process which creates instances is just selecting a random row from the given dataset
- Under these assumptions we can calculate probabilities by counting the proportion of instances that satisfy the respective condition

# Tennis dataset

Does there exist a classifier which is always correct on this dataset?

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Optimal classifier on training data?

- The perfect classifier does not exist on Tennis dataset:  
( $x=\text{Rain}, y=\text{Yes}$ ) and ( $x=\text{Rain}, y=\text{No}$ ) are both in the data but any classifier maps Rain to either Yes or No
- Remember that classifier is just a function of features:  $\hat{y} = f(\mathbf{x})$
- Which is the optimal classifier on training data? In the sense that it makes the fewest errors

# Optimal classifier on training data

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
| D7  | Overcast | Yes        |
| D12 | Overcast | Yes        |
| D13 | Overcast | Yes        |
| D4  | Rain     | Yes        |
| D5  | Rain     | Yes        |
| D6  | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
| D9  | Sunny    | Yes        |
| D11 | Sunny    | Yes        |

# Optimal classifier on training data

Optimal classifier on these data:

- $f(Overcast) = Yes$
- $f(Rain) = Yes$
- $f(Sunny) = No$

| Day | Outlook  | PlayTennis |
|-----|----------|------------|
| D3  | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Overcast | Yes        |
|     | Rain     | Yes        |
|     | Rain     | Yes        |
|     | Rain     | No         |
| D10 | Rain     | Yes        |
| D14 | Rain     | No         |
| D1  | Sunny    | No         |
| D2  | Sunny    | No         |
| D8  | Sunny    | No         |
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# Optimal classifier on training data

- In general, how to obtain the optimal classifier?

$$\hat{y} = f(\mathbf{x}) = ?$$

# Optimal classifier on training data

- In general, how to obtain the optimal classifier?

$$\hat{y} = f(\mathbf{x}) = ?$$

- Proportion of rows with feature values  $\mathbf{x}$  is:

$$P(\mathbf{X} = \mathbf{x})$$

# Optimal classifier on training data

- In general, how to obtain the optimal classifier?

$$\hat{y} = f(\mathbf{x}) = ?$$

- Proportion of rows with feature values  $\mathbf{x}$  is:

$$P(\mathbf{X} = \mathbf{x})$$

- Among those, proportion of positives:

$$P(Y = \oplus | \mathbf{X} = \mathbf{x})$$

- Among those, proportion of negatives:

$$P(Y = \ominus | \mathbf{X} = \mathbf{x})$$

# Optimal classifier on training data

- If our model predicts positive:  $\hat{y} = f(\mathbf{x}) = \oplus$   
Then it makes mistakes on the negatives:

$$P(Y = \ominus | \mathbf{X} = \mathbf{x})$$

# Optimal classifier on training data

- If our model predicts positive:  $\hat{y} = f(\mathbf{x}) = \oplus$   
Then it makes mistakes on the negatives:

$$P(Y = \ominus | \mathbf{X} = \mathbf{x})$$

- If our model predicts negative:  $\hat{y} = f(\mathbf{x}) = \ominus$   
Then it makes mistakes on the positives:

$$P(Y = \oplus | \mathbf{X} = \mathbf{x})$$

# Optimal classifier on training data

- If our model predicts positive:  $\hat{y} = f(\mathbf{x}) = \oplus$

Then it makes mistakes on the negatives:

$$P(Y = \ominus | \mathbf{X} = \mathbf{x})$$

- If our model predicts negative:  $\hat{y} = f(\mathbf{x}) = \ominus$

Then it makes mistakes on the positives:

$$P(Y = \oplus | \mathbf{X} = \mathbf{x})$$

- To minimize errors we should:
  - Predict  $\oplus$  if  $P(Y = \oplus | \mathbf{X} = \mathbf{x}) > P(Y = \ominus | \mathbf{X} = \mathbf{x})$
  - Predict  $\ominus$  if  $P(Y = \ominus | \mathbf{X} = \mathbf{x}) > P(Y = \oplus | \mathbf{X} = \mathbf{x})$

# Maximum a posteriori (MAP)

Optimal is to follow the **maximum a posteriori (MAP)** decision rule

$$\hat{y} = f(\mathbf{x}) = \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y | \mathbf{X} = \mathbf{x})$$

That is, predict the class that has the highest probability conditional to the given feature values

- If we somehow magically know probabilities according to the true process that generates instances, then MAP-decision rule gives us **Bayes classifier**
- Bayes classifier is defined by:

$$\hat{y} = f(\mathbf{x}) = \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y | \mathbf{X} = \mathbf{x}),$$

where  $P$  is the probability measure defined by the true data-generating process

# Have we solved classification?

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- We have just found how to calculate the optimal classifier – have we solved classification?
- **No!** We made the assumption that instances are drawn randomly from the training data
- When testing our classifier on new test data, we might encounter a different distribution of data and even new instances

# Train and Test data

- So far we have obtained the optimal classifier on our training data, but having a good classifier on training data is not sufficient
- The goal is to perform well on future data
- How well would our classifier perform on future data?
  - We will never know exactly in advance, but we can estimate it by applying the classifier on a test dataset, which is separate from the training dataset
- What if we don't have test data?
  - We can still estimate how well our learning algorithm works by splitting the original training data randomly into training and test data

# Random split into train and test sets

| Day | Outlook  | PlayTennis | T<br>r<br>a<br>i<br>n<br>i<br>n<br>g |
|-----|----------|------------|--------------------------------------|
| D1  | Sunny    | No         |                                      |
| D2  | Sunny    | No         |                                      |
| D4  | Rain     | Yes        |                                      |
| D6  | Rain     | No         |                                      |
| D7  | Overcast | Yes        |                                      |
| D12 | Overcast | Yes        |                                      |
| D14 | Rain     | No         |                                      |
|     |          |            | T<br>e<br>s<br>t<br>i<br>n<br>g      |
| D3  | Overcast | Yes        |                                      |
| D5  | Rain     | Yes        |                                      |
| D8  | Sunny    | No         |                                      |
| D9  | Sunny    | Yes        |                                      |
| D10 | Rain     | Yes        |                                      |
| D11 | Sunny    | Yes        |                                      |
| D13 | Overcast | Yes        |                                      |

# Random split into train and test sets

| Day | Outlook  | PlayTennis | T<br>r<br>a<br>i<br>n<br>i<br>n<br>g |
|-----|----------|------------|--------------------------------------|
| D1  | Sunny    | No         |                                      |
| D2  | Sunny    | No         |                                      |
| D4  | Rain     | Yes        |                                      |
| D6  | Rain     | No         |                                      |
| D7  | Overcast | Yes        |                                      |
| D12 | Overcast | Yes        |                                      |
| D14 | Rain     | No         |                                      |
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| D3  | Overcast | Yes        |                                      |
| D5  | Rain     | Yes        |                                      |
| D8  | Sunny    | No         |                                      |
| D9  | Sunny    | Yes        |                                      |
| D10 | Rain     | Yes        |                                      |
| D11 | Sunny    | Yes        |                                      |
| D13 | Overcast | Yes        |                                      |

Optimal classifier

on training data:

$$f(Overcast) = Yes$$

$$f(Rain) = No$$

$$f(Sunny) = No$$

Train-optimal classifier

makes 4 errors:

D5,D9,D10,D11

# Bad performance on test data

- Our train-optimal classifier had 4/7 error rate on the test data – why so bad?
  - We might have been unlucky with how the training and test data were split
  - The dataset is small - a bigger dataset might have allowed us to learn more successfully

# Bad performance on test data

- Our train-optimal classifier had  $4/7$  error rate on the test data – why so bad?
  - We might have been unlucky with how the training and test data were split
  - The dataset is small - a bigger dataset might have allowed us to learn more successfully

Let's add another feature from the initial dataset

# We have 2 features now

| Day | Outlook  | Temp | PlayTennis |
|-----|----------|------|------------|
| D7  | Overcast | Cool | Yes        |
| D3  | Overcast | Hot  | Yes        |
| D13 | Overcast | Hot  | Yes        |
| D12 | Overcast | Mild | Yes        |
| D5  | Rain     | Cool | Yes        |
| D6  | Rain     | Cool | No         |
| D4  | Rain     | Mild | Yes        |
| D10 | Rain     | Mild | Yes        |
| D14 | Rain     | Mild | No         |
| D9  | Sunny    | Cool | Yes        |
| D1  | Sunny    | Hot  | No         |
| D2  | Sunny    | Hot  | No         |
| D8  | Sunny    | Mild | No         |
| D11 | Sunny    | Mild | Yes        |

## We have 2 features now

| Day | Outlook  | Temp | PlayTennis |
|-----|----------|------|------------|
| D7  | Overcast | Cool | Yes        |
| D3  | Overcast | Hot  | Yes        |
| D13 | Overcast | Hot  | Yes        |
| D12 | Overcast | Mild | Yes        |
| D5  | Rain     | Cool | Yes        |
| D6  | Rain     | Cool | No         |
| D4  | Rain     | Mild | Yes        |
| D10 | Rain     | Mild | Yes        |
| D14 | Rain     | Mild | No         |
| D9  | Sunny    | Cool | Yes        |
| D1  | Sunny    | Hot  | No         |
| D2  | Sunny    | Hot  | No         |
| D8  | Sunny    | Mild | No         |
| D11 | Sunny    | Mild | Yes        |

Optimal classifier

on these data:

$f(\text{Overcast}, \text{Cool}) = \text{Yes}$

$f(\text{Overcast}, \text{Hot}) = \text{Yes}$

$f(\text{Overcast}, \text{Mild}) = \text{Yes}$

$f(\text{Rain}, \text{Cool}) = \text{Yes/No}$

$f(\text{Rain}, \text{Mild}) = \text{Yes}$

$f(\text{Sunny}, \text{Cool}) = \text{Yes}$

$f(\text{Sunny}, \text{Hot}) = \text{No}$

$f(\text{Sunny}, \text{Mild}) = \text{Yes/No}$

## We have 2 features now

| Day | Outlook  | Temp | PlayTennis |
|-----|----------|------|------------|
| D7  | Overcast | Cool | Yes        |
| D3  | Overcast | Hot  | Yes        |
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| D4  | Rain     | Mild | Yes        |
| D10 | Rain     | Mild | Yes        |
| D14 | Rain     | Mild | No         |
| D9  | Sunny    | Cool | Yes        |
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Optimal classifier

on these data:

$$f(\text{Overcast}, \text{Cool}) = \text{Yes}$$

$$f(\text{Overcast}, \text{Hot}) = \text{Yes}$$

$$f(\text{Overcast}, \text{Mild}) = \text{Yes}$$

$$f(\text{Rain}, \text{Cool}) = \text{Yes/No}$$

$$f(\text{Rain}, \text{Mild}) = \text{Yes}$$

$$f(\text{Sunny}, \text{Cool}) = \text{Yes}$$

$$f(\text{Sunny}, \text{Hot}) = \text{No}$$

$$f(\text{Sunny}, \text{Mild}) = \text{Yes/No}$$

$$f(\text{Rain}, \text{Hot}) = ?$$

# Problem with train-optimal classifier

- Our train-optimal classifier does not define what to predict for  $f(Rain, Hot)$
- If an instance with these feature values occurs in future data then we would need to predict randomly
- The same problem gets worse with more features

# Tennis dataset with all 4 features

| Day | Outlook  | Temp | Humidity | Wind   | PlayTennis |
|-----|----------|------|----------|--------|------------|
| D1  | Sunny    | Hot  | High     | Weak   | No         |
| D2  | Sunny    | Hot  | High     | Strong | No         |
| D3  | Overcast | Hot  | High     | Weak   | Yes        |
| D4  | Rain     | Mild | High     | Weak   | Yes        |
| D5  | Rain     | Cool | Normal   | Weak   | Yes        |
| D6  | Rain     | Cool | Normal   | Strong | No         |
| D7  | Overcast | Cool | Normal   | Strong | Yes        |
| D8  | Sunny    | Mild | High     | Weak   | No         |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes        |
| D10 | Rain     | Mild | Normal   | Weak   | Yes        |
| D11 | Sunny    | Mild | Normal   | Strong | Yes        |
| D12 | Overcast | Mild | High     | Strong | Yes        |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes        |
| D14 | Rain     | Mild | High     | Strong | No         |

# Tennis dataset with all 4 features

| Day                                                                                                     | Outlook                                                                                                                                                                                                                       | Temp | Humidity | Wind   | PlayTennis |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|--------|------------|
| D1                                                                                                      | Sunny                                                                                                                                                                                                                         | Hot  | High     | Weak   | No         |
| D2                                                                                                      | Sunny                                                                                                                                                                                                                         | Hot  | High     | Strong | No         |
| D3                                                                                                      | Overcast                                                                                                                                                                                                                      | Hot  | High     | Weak   | Yes        |
| D4                                                                                                      | 14 instances but $3 \times 3 \times 2 \times 2 = 36$<br>different possible feature value<br>combinations:<br>$\{Overcast, Rain, Sunny\} \times$<br>$\{Cool, Hot, Mild\} \times$<br>$\{High, Normal\} \times \{Strong, Weak\}$ |      |          |        | Yes        |
| D5                                                                                                      |                                                                                                                                                                                                                               |      |          |        | Yes        |
| D6                                                                                                      |                                                                                                                                                                                                                               |      |          |        | No         |
| D7                                                                                                      |                                                                                                                                                                                                                               |      |          |        | Yes        |
| D8                                                                                                      |                                                                                                                                                                                                                               |      |          |        | No         |
| D9                                                                                                      |                                                                                                                                                                                                                               |      |          |        | Yes        |
| D10                                                                                                     | Rain                                                                                                                                                                                                                          | Mild | Normal   | Weak   | Yes        |
| D11                                                                                                     | Sunny                                                                                                                                                                                                                         | Mild | Normal   | Strong | Yes        |
| Therefore, we would need to predict randomly on<br>most of the feature value combinations in the future |                                                                                                                                                                                                                               |      |          |        |            |
| D14                                                                                                     | Rain                                                                                                                                                                                                                          | Mild | High     | Strong | No         |

# Give up exact feature combinations

- We need to give up on expecting to see the same feature combinations in train and test data
- What to do if we have a test instance with an unseen feature combination?
- We will see two solutions soon:
  - K Nearest Neighbours
  - Naïve Bayes

| Day | Outlook  | Temp | Humidity | Wind   | PlayTennis |
|-----|----------|------|----------|--------|------------|
| D1  | Sunny    | Hot  | High     | Weak   | No         |
| D2  | Sunny    | Hot  | High     | Strong | No         |
| D3  | Overcast | Hot  | High     | Weak   | Yes        |
| D4  | Rain     | Mild | High     | Weak   | Yes        |
| D5  | Rain     | Cool | Normal   | Weak   | Yes        |

## Question

What would you predict for  $\mathbf{x} = (\text{Sunny}, \text{Hot}, \text{Normal}, \text{Strong})$ ?

# 1 Nearest Neighbour (1NN)

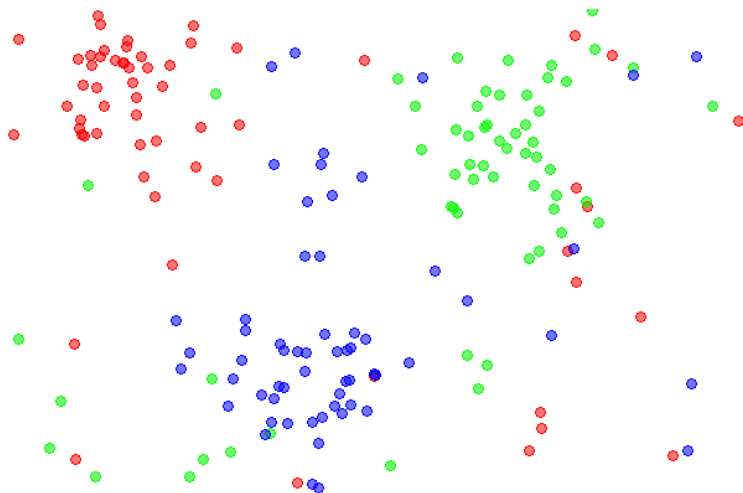
- It is natural to hope that if two instances differ by only 1 feature then they are very likely to belong to the same class
- 1 nearest neighbour algorithm (1NN):
  - For a given test instance  $\mathbf{x}$
  - Find the training instance  $\mathbf{x}_i$  that is nearest to  $\mathbf{x}$
  - Predict the label  $\mathbf{y}_i$  on that training instance
- Need to define what we mean by "*nearest to*"
  - For categorical features – we could find the instance with the most shared feature values

# K-Nearest Neighbours (KNN)

- **Pros:** One of the simplest learning methods, applies for both categorical and numerical data
- $K$  - integer, usually an odd number (1, 3, 5, ...)
- Learning the model=Memorise training data
- Applying the model on test data:
  - For each test instance:
    - Find the  $K$  closest training instances
    - Predict most frequent label (in classification) or average label (in regression) among those  $K$
- **Cons:** hard to choose a good distance measure and many distance calculations are required

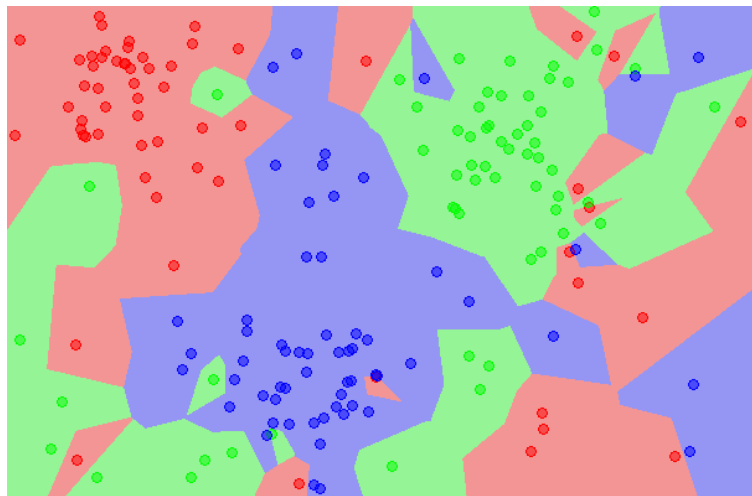
# KNN example in 2D

Figure: Data points in 2D



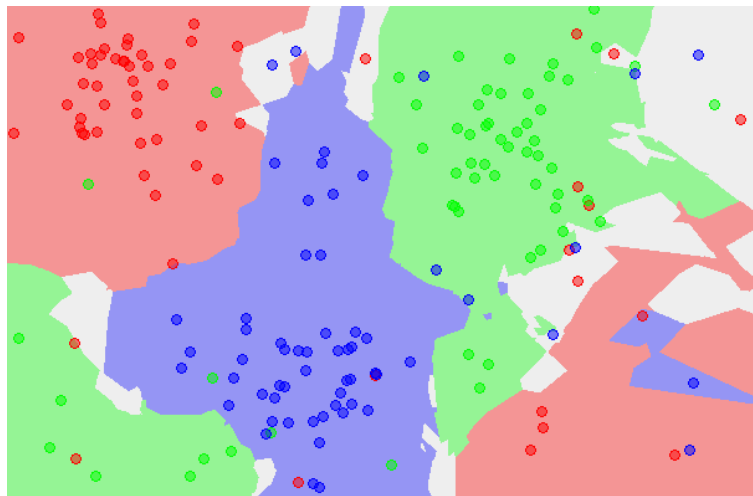
# KNN example in 2D

Figure: Classification regions based on 1NN



# KNN example in 2D

Figure: Classification regions based on 5NN



# Well-known distance measure

- For numeric vectors  $\mathbf{x} = (x_1, \dots, x_m)$  and  $\mathbf{x}' = (x'_1, \dots, x'_m)$

- Euclidean distance:  $d(\mathbf{x}, \mathbf{x}') = \sqrt{\sum_{i=1}^m (x_i - x'_i)^2}$

- Manhattan distance:  $d(\mathbf{x}, \mathbf{x}') = \sum_{i=1}^m |x_i - x'_i|$

- Correlation distance:  $d(\mathbf{x}, \mathbf{x}') = 1 - \rho(\mathbf{x}, \mathbf{x}')$ , where  $\rho(\mathbf{x}, \mathbf{x}')$  is Pearson correlation coefficient

- For categorical vectors

- Hamming distance:  $d(\mathbf{x}, \mathbf{x}') = \sum_{i=1}^m \mathbb{1}_{\{x_i \neq x'_i\}}$ , where  $\mathbb{1}_{\{\cdot\}}$  is the indicator function.

What to do if we have a test instance with an unseen feature combination?

- KNN provided one possible solution
- Let us look for a probabilistic solution

- Conditional Probability:  $P(A|B) = \frac{P(A, B)}{P(B)}$
- Chain rule:  $P(A, B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$
- Bayes' rule:  $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- If A, B are independent ( $A \perp B$ ), then  
 $P(A, B) = P(A) \cdot P(B)$  and  $P(A|B) = P(A)$ ,  $P(B|A) = P(B)$

## Definition: Conditional independence

If A, B are conditionally independent given C (we will denote  $A \perp B|C$ ), then

$$P(A, B|C) = P(A|C) \cdot P(B|C)$$

and

$$P(A|B, C) = P(A|C); \quad P(B|A, C) = P(B|C)$$

- Assume that any pair of features are conditionally independent given the label:

$$X_i \perp X_j | Y, \quad i \neq j$$

- Then MAP decision rule gives us:

$$\hat{y} = f(\mathbf{x}) = \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y | \mathbf{X} = \mathbf{x}) =$$

- Assume that any pair of features are conditionally independent given the label:

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- Then MAP decision rule gives us:

$$\begin{aligned}\hat{y} = f(\mathbf{x}) &= \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y | \mathbf{X} = \mathbf{x}) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(\mathbf{X} = \mathbf{x} | Y = y) \cdot P(Y = y) / P(\mathbf{X} = \mathbf{x}) =\end{aligned}$$

- Assume that any pair of features are conditionally independent given the label:

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- Assume that any pair of features are conditionally independent given the label:

$$X_i \perp X_j | Y, \quad i \neq j$$

- Then MAP decision rule gives us:

$$\begin{aligned}\hat{y} = f(\mathbf{x}) &= \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y | \mathbf{X} = \mathbf{x}) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(\mathbf{X} = \mathbf{x} | Y = y) \cdot P(Y = y) / P(\mathbf{X} = \mathbf{x}) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(\mathbf{X} = \mathbf{x} | Y = y) \cdot P(Y = y) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(X_1 = x_1, \dots, X_m = x_m | Y = y) \cdot P(Y = y) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(X_1 = x_1 | Y = y) \dots P(X_m = x_m | Y = y) \cdot P(Y = y) =\end{aligned}$$

# Naïve Bayes

- Assume that any pair of features are conditionally independent given the label:

$$X_i \perp X_j | Y, \quad i \neq j$$

- Then MAP decision rule gives us:

$$\begin{aligned}\hat{y} = f(\mathbf{x}) &= \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y | \mathbf{X} = \mathbf{x}) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(\mathbf{X} = \mathbf{x} | Y = y) \cdot P(Y = y) / P(\mathbf{X} = \mathbf{x}) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(\mathbf{X} = \mathbf{x} | Y = y) \cdot P(Y = y) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(X_1 = x_1, \dots, X_m = x_m | Y = y) \cdot P(Y = y) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(X_1 = x_1 | Y = y) \dots P(X_m = x_m | Y = y) \cdot P(Y = y) = \\ &= \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y) \prod_{i=1}^m P(X_i = x_i | Y = y)\end{aligned}$$

# Naïve Bayes

- Assume that any pair of features are conditionally independent given the label:

$$X_i \perp X_j | Y, \quad i \neq j$$

- Then MAP decision rule gives us:

$$\hat{y} = f(\mathbf{x}) = \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y | \mathbf{X} = \mathbf{x}) =$$

Naïve Bayes

= Predict the class according to the following rule:

$$= \hat{y} = f(\mathbf{x}) = \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y) \prod_{i=1}^m P(X_i = x_i | Y = y)$$

$$= \operatorname{argmax}_{y \in \mathbb{Y}} P(X_1 = x_1 | Y = y) \dots P(X_m = x_m | Y = y) \cdot P(Y = y) =$$

$$= \operatorname{argmax}_{y \in \mathbb{Y}} P(Y = y) \prod_{i=1}^m P(X_i = x_i | Y = y)$$

# Example

What would Naïve Bayes predict for  $x = (\textit{Sunny}, \textit{Hot}, \textit{Normal}, \textit{Strong})$ ?

| Day | Outlook  | Temp | Humidity | Wind   | PlayTennis |
|-----|----------|------|----------|--------|------------|
| D1  | Sunny    | Hot  | High     | Weak   | No         |
| D2  | Sunny    | Hot  | High     | Strong | No         |
| D3  | Overcast | Hot  | High     | Weak   | Yes        |
| D4  | Rain     | Mild | High     | Weak   | Yes        |
| D5  | Rain     | Cool | Normal   | Weak   | Yes        |
| D6  | Rain     | Cool | Normal   | Strong | No         |
| D7  | Overcast | Cool | Normal   | Strong | Yes        |
| D8  | Sunny    | Mild | High     | Weak   | No         |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes        |
| D10 | Rain     | Mild | Normal   | Weak   | Yes        |
| D11 | Sunny    | Mild | Normal   | Strong | Yes        |
| D12 | Overcast | Mild | High     | Strong | Yes        |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes        |
| D14 | Rain     | Mild | High     | Strong | No         |

# Example

- What would Naïve Bayes predict for  $x = (\text{Sunny}, \text{Hot}, \text{Normal}, \text{Strong})$ ?

$$P(Y = \oplus) \prod_{i=1}^m P(X_i = x_i | Y = \oplus) = \frac{9}{14} \cdot \frac{2}{9} \cdot \frac{2}{9} \cdot \frac{6}{9} \cdot \frac{3}{9} = \frac{4}{567} \approx 0.007$$

$$P(Y = \ominus) \prod_{i=1}^m P(X_i = x_i | Y = \ominus) = \frac{5}{14} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} \cdot \frac{3}{5} = \frac{9}{875} \approx 0.01$$

- Naïve Bayes predicts negative because:

$$P(Y = \ominus) \prod_{i=1}^m P(X_i = x_i | Y = \ominus) > P(Y = \oplus) \prod_{i=1}^m P(X_i = x_i | Y = \oplus)$$

- When calculating probabilities in

$$P(Y = y) \prod_{i=1}^m P(X_i = x_i | Y = y)$$

any zero probability would make the total into zero also

- Usually, **Laplace smoothing** is applied when calculating these probabilities, so instead of dividing  $\frac{a}{b}$ , where  $a$  are favourable and  $b$  are all options, in Laplace smoothing one would use the ratio  $\frac{a+1}{b+k}$ , where  $k$  is the number of unique values of the given feature

# Example with Laplace smoothing

- What would Naïve Bayes predict for  $x = (\text{Sunny}, \text{Hot}, \text{Normal}, \text{Strong})$ ?

$$P(Y = \oplus) \prod_{i=1}^m P(X_i = x_i | Y = \oplus) = \frac{9}{14} \cdot \frac{2 + \mathbf{1}}{9 + \mathbf{3}} \cdot \frac{2 + \mathbf{1}}{9 + \mathbf{3}} \cdot \frac{6 + \mathbf{1}}{9 + \mathbf{2}} \cdot \frac{3 + \mathbf{1}}{9 + \mathbf{2}} \approx 0.0093$$

$$P(Y = \ominus) \prod_{i=1}^m P(X_i = x_i | Y = \ominus) = \frac{5}{14} \cdot \frac{3 + \mathbf{1}}{5 + \mathbf{3}} \cdot \frac{2 + \mathbf{1}}{5 + \mathbf{3}} \cdot \frac{1 + \mathbf{1}}{5 + \mathbf{2}} \cdot \frac{3 + \mathbf{1}}{5 + \mathbf{2}} \approx 0.01$$

- Again, Naïve Bayes predicts negative because:

$$P(Y = \ominus) \prod_{i=1}^m P(X_i = x_i | Y = \ominus) > P(Y = \oplus) \prod_{i=1}^m P(X_i = x_i | Y = \oplus)$$

- **Pro:** simple and takes into account the class prior
- **Con:** assumes conditional independence of features given the label

- **Pro:** simple and takes into account the class prior
- **Con:** assumes conditional independence of features given the label
- Being "Naïve" can be a good thing:

Occam's razor (important guideline in machine learning)

Everything should be made as simple as possible, but not simpler.

# What does "Bayesian" mean?

- It means that you have some prior belief about the probability of some event(s) before observing the data
- You update your belief according to the likelihood of new observations
- In our case:
  - Prior belief was  $P(Y = y)$
  - We observed  $\mathbf{X} = \mathbf{x}$
  - Likelihood of observations:  $P(\mathbf{X} = \mathbf{x} | Y = y)$
  - Posterior (updated) belief:

$$P(Y = y | \mathbf{X} = \mathbf{x}) = \frac{P(Y = y) \cdot P(\mathbf{X} = \mathbf{x} | Y = y)}{P(\mathbf{X} = \mathbf{x})}$$

# What have we learned today?

- ✓ K-Nearest Neighbours Classifier
- ✓ Naive Bayes Classifier